

Traditional uses, processes, and markets: the case of achira (*Canna indica* L.)

Nelly Judith Paredes Andrade¹, Juan Pablo Rodríguez² and
Marten Sørensen³

¹National Agricultural Research Institute (INIAP), Central Experimental Station of the Amazon, Joya de los Sachas, Orellana, Ecuador ²Julius Kühn-Institut (JKI) - Federal Research Centre for Cultivated Plants, Institute for Plant Protection in Horticulture and Urban Green, Braunschweig, Germany ³Department of Plant and Environmental Sciences, University of Copenhagen, Frederiksberg C, Copenhagen, Denmark

21.1 Origin

The achira comes from the Quechua word *achuy*. It is also known in the Andes as Peruvian cane, Indian cane, Mexican railing, crab flower, red plantain, *yerba del rosario*, *sagú*, *chisgua*, *capacho*, *maraca*, *zulu*, *pericaguá*, *achera*, *atcera*, *biri*, *acaxochitl*, *papantla*, *cucuyús*, *juquián*, *cañacoro*, *chui*, *arawak*, *imocoma*, *chiri*, *luano*, *canna*, and *chuno* (Gade, 1966; Tapia & Fries, 2007; Ruilova, 2013). It is also known in many languages, such as Arabic (*canna* Hindi, *muzwardi*, and *muzfahal*), English (*canna* Indian shot, *achira*, and *arrowroot*), French (*balisier*), Hindi (*sakasiri* and *devkali*), Marathi (*kardal*), and Sanskrit (*vankelee*). In Australia it is named Queensland arrowroot; in Taiwan, *lotus*; in Thailand, *sagú*; and in Spanish, *chupaflor* (Lobo-Arias et al., 2016; Imai, 2008).

Canna indica, the achira used for starch extraction, belongs to the Cannaceae family (Tanaka,

2004). It was classified as *C. indica* by Linnaeus in 1753 and as *Canna edulis* by Ker Gawler in 1824, names that are considered synonymous (Tropicos, 2016; PoWo (Plants of the World Online), 2024; WFO (World Flora Online), 2024). The Neotropical genus *Canna* L. encompasses 12 species and 1 natural hybrid: *Canna bangii* Kraenzl., *Canna flaccida* Salisb., *Canna glauca* L., *C. indica* L., *Canna iridiflora* Ruiz & Pav., *Canna jaeegeriana* Urb., *Canna liliiflora* Warsz. ex Planch., *Canna lineata* Ciciar., *Canna paniculata* Ruiz & Pav., *Canna pedunculata* Sims, *Canna tandilensis* Ciciar., *Canna tuerckheimii* Kraenzl., and the natural hybrid *Canna* × *hybrida* Rodigas (Caicedo et al., 2000; PoWo (Plants of the World Online), 2024; WFO (World Flora Online), 2024).

The achira originated in South America. In Peru, archeological remains were found, such as vessels molded in the shape of achira rhizomes belonging to the Chimú and Nazca cultures, dating back 2500 years BCE (Gade, 1966).

However, it is not considered a wild plant. The center of domestication is considered to be the tropical area of Colombia, from where it may have spread through the Andes to the western coast to northern Chile. In this way, we can find it throughout tropical America (Morales Ordoñez, 2016; Hernández Sierra, 2017; Gawl et al., 2017; Flórez Londoño & Valencia Correa, 2019). It is grown in many regions of the world to produce starch, such as in Colombia, Ecuador, Peru, Bolivia, Venezuela, Brazil, Mexico, and northern Chile. It has been introduced in Taiwan, Vietnam, Thailand, India, China, Oceania, Asia, Africa, and Polynesia since the rhizomes have a high starch content (Piyachomkwan et al., 2002; Caicedo, 2004). For information on extracting starch from achira and producing pasta, we recommend Chapters 3 and 8 in the Third Volume of this series.

The agroecological conditions range from 1000 to 2900 m a.s.l., with the most appropriate altitude for cultivation being between 2000 and 2650 m a.s.l. But achira crops have also been observed at sea level. It is a neutral photoperiod plant, that is, it is not affected by the length of the hours of darkness. It supports temperatures between 16°C and 32°C, the optimal temperature being between 18°C and 24°C, and rainfall from 250 to 4000 mm per year. In Taiwan, Vietnam, and southern China,

it develops between 2000 and 3000 m a.s.l. (Roca & Manrique, 2005; Reichel, 2007; Zhang et al., 2010).

Achira is a crop managed traditionally without agrochemicals since it does not present attacks by pests or diseases. It grows in different types of soils, the ideal being loose, clayey, sandy clay, or silty clay soils and rich in organic matter. In periods of maturation of the rhizomes, the achira does not tolerate frost. Fig. 21.1 shows a recent planting of achira, with a participatory evaluation by producers (A) and a flowering field of achira with a flower in focus (B).

It is a monocotyledonous perennial plant with a fleshy pseudo-stem with several large, underground, and tuber-shaped rhizomes, 1.5–3.0 m high. Phenotypically, green achiras are distinguished, with white rhizomes and green leaves, and purple achiras, with rhizomes with purple coverage and purple leaves. The stems are in the shape of clusters, their leaves have a thick central vein that can measure up to 85 cm long, with a simple or branched terminal inflorescence in the form of a loose raceme. The flowers are hermaphrodite, red or yellow, with fruits in brown capsules (Caicedo, 2004; Pazto Pasto et al., 2013).

The harvest is carried out manually by uprooting the plant with a shovel, bar, or hoe.



FIGURE 21.1 (A) Field with achira at the beginning of the growth phase, with a participatory evaluation by producers, and (B) a flowering field of achira with a flower highlighted in the right-hand corner. Source: Photo A from INIAP & J. Romero; Photo B from J. Romero.

The particles of soil are removed, and the stem is cut to separate the rhizomes and processed immediately after harvest, for starch extraction (Ariza-León et al., 2013; Punched-arnon et al., 2007; Caguana Telenchana, 2018; Flórez Londoño & Valencia Correa, 2019) (Fig. 21.2).

Predominant color of the achira flower is red; however, rhizome color may vary, for example, the three cultivars tested in Colombia by Caicedo et al. (2020): first cultivar with purple rhizome from municipality of San Agustín, second white achira cultivar from municipality of Cruz (Nariño), and the native cultivar from municipality of Altamira. Achira purple rhizome cultivar gave a high yield with $18.8 \text{ t} \cdot \text{ha}^{-1}$.

The extraction of achira starch is an essential generator of economic resources. The starch is obtained from the plant's rhizomes, characterized by its considerable rapidity of sedimentation, having a large particle diameter, of 30–100 microns (Torres, 2004). The amylose content of achira starch is high and the grain size is large compared to maize and potato. Cisneros et al. (2009) analyzed two cultivars of Peruvian achira—the achira Sandia had 33.1%, and achira San Gabán had 39.4% amylose content, respectively, while maize and potato had 23.8% and 27%, respectively. Thus, when achira has a high amylose content, it is a starch with a high consistency gel property. For more



FIGURE 21.2 Achira rhizome (A), (B) harvesting the achira and separating the rhizomes, (C) cleaning the rhizomes, and (D) bags for transporting the achira rhizomes from the field to the market. Source: *Photos from J. Romero.*

information on the properties of achira starch and other underground starchy crops, see Chapters 8–10 in the First Volume of this series.

Due to its leaf mass production, achira provides environmental benefits (Fig. 21.3). At the time of harvest, it has an average of $21 \text{ t} \cdot \text{ha}^{-1}$ of biomass, which improves and enriches the soil in organic matter. It should be mentioned that there are ecotypes and species for ornamental, industrial, or wild use (Cardenas, 1950).

The rhizomes of the achira are generally used for starch extraction. The artisanal process to obtain starch consists of grating the rhizomes (Fig. 21.4A), then adding water and proceeding to filtration (Fig. 21.4B), allowing the separation of the starch particles suspended in the liquid medium. Subsequently, the water is removed by decantation (Fig. 21.4C), and finally, the filtered starch is dried (Alarcón & Dufour, 1998, see also Chapter 3 in the Third Volume of this series). Because it contains a lot

of phenols, the settling water turns black, and the starch needs to be washed several times.

Achira starch can be used in the food, pharmaceutical, beverage, and sizing industries, among others. Since it is considered as easily digestible, it can be used as food for children, older adults, people with some diseases, and people with digestive problems (Hermann, 1994).

21.2 Traditional uses—medicinal, ceremonial, etc

Since the time of the Incas, achira starch has been used in food. Therefore starch is a heritage of the aborigines as it is inserted into the gastronomic tradition, becoming a cultural heritage, and contributing to food sovereignty. They consumed it parboiled as a garnish and called this preparation *San*. In the same way, the corms of the achira were also consumed roasted or cooked.



FIGURE 21.3 Experimental cultivation of achira in central Brazil, at an altitude of 500 m a.s.l. and with plants over 2 m tall (A) and harvest of achira in China with significant aerial part remaining (B). Source: Photos from O.F. Vilpoux.



FIGURE 21.4 Grating the achira (A), extracting the starch through a sieve and adding water (B), decanting achira starch, still containing phenols, and requiring additional washing (C) and achira starch (D). Source: Photos A–E from J. Romero; Photo F from D. Freire.

The starch was also used in the preparation of *coladas*, *arepas*, homemade bread, especially in artisanal bakeries to produce biscuit, sago bread, snacks, curds, *mazamorras*, cookies, *kekes*, and snacks (Fig. 21.5). On the other hand, the achira leaf is used for wrapping traditional foods like *tamales* and *quimbolitos*. Currently the rhizomes are processed in artisanal factories on small family farms, to obtain starch

(Piyachomkwan et al., 2002; Caicedo et al., 2003; Lares & Pérez, 2006; Gawl et al., 2017).

In Asian regions such as Vietnam and China, the rhizomes are transformed into starch for the preparation of noodles, a traditional oriental food from Southeast Asia (Fig. 21.6A) (Montaldo, 1972; Roca & Manrique, 2005; Imai, 2008; Ruilova, 2013; Andrade-Mahecha et al., 2012; Chapter 8 in the Third Volume of this series).



FIGURE 21.5 (A) Achira milk, (B) achira *colada* (achira strain), (C) achira bread, (D) achira cookies, (E) achira cakes, (F) achira and roses brownie, (G) achira muffin, (H) cheesecake with achira starch, (I) achira and bananas cake, (J) achira *alfajores* (soft shortbread cookies with a creamy *dulce de leche*), and (K) achira pancakes. Source: Photos A and B from J. Romero; Photos C and K from D. Freire; Photos E–H from K. Morales.

Achira fiber has been reported for use in dietary foods and additives in the food industry, due to its high amylose content (Hermann et al., 1999; Kasemwong et al., 2003; Tanaka et al., 2023). Starch is also used as a thickener in instant soups, in the manufacture of sauces, diet products, sweets, and gums. Rhizomes can also be eaten raw or cooked like potatoes (Fig. 21.6B).

The rhizomes of the achira are used to make tea since it is considered a diuretic. The juice of achira is used as an antiseptic, and the freshly cut leaves are used on burns to cool and dissipate the heat of the burned skin and as healing agents. They are also used as a type of natural packaging to wrap typical foods such as *tamales*. The seeds are used to make necklaces

and rattles or maracas. The stems and leaves serve as fodder for livestock. Rhizomes are slightly sweet and are consumed cooked, roasted, pureed or fried, in soups, stews, and fried in slices. The foliage is used in the preparation of raw or cooked salads in the style of celery (Zhang et al., 2010).

The achira plant has pharmacological properties with several therapeutic effects since it contains flavonoids, tannins, saponins, botulinic acid, oleanolic acid, lutein, and β -carotenes. They are present in the leaves, stems, rhizomes, and seeds, so that they can be used as an antibacterial, antiviral, anthelmintic, analgesic, antioxidant, hemostatic, hepatoprotective, and antidiabetic product (Zhang et al., 2011; Yaruro Cáceres, 2018).



FIGURE 21.6 (A) Drying of achira noodles in China, and (B) commercial achira sold in Australia for production of edible rhizomes. Source: Photo A from O.F. Vilpoux; Photo B from M. Sørensen.

According to Sarje et al. (2019), achira has been widely used in traditional medicine for the treatment of many complaints. Phytochemical analysis of achira showed that it contained various phytochemicals, including flavonoids, terpenoids, alkaloids, steroids, cardiac glycosides, oils, saponins, tannins, anthocyanin pigments, phlobatinins and many other chemical compounds. Pharmacological studies have shown that the achira plant has antiviral, anthelmintic, antiinflammatory, antibacterial, antioxidant, molluscicidal, cytotoxic, hepatoprotective, analgesic, immunomodulatory, hemostatic, antidiarrheal, and other effects. For the authors, each part of *Canna* has beneficial properties that can serve humanity.

It is possible to find some nonconventional uses in Thailand, where achira is used to treat

cancer—effects studied by Otari et al. (2017) and Burhannudin et al. (2018); in the Southwest of Nigeria, where it is used to cure malaria and treat hepatitis; and in the Philippines, where rhizome tea is used for its diuretic effect and the maceration of the leaves to stop nosebleeds. In Costa Rica, the infusion of the leaves is taken to relieve stomach cramps, and in Brazil, the infusion of the leaves and rhizomes is used to clean ulcers and the steam to treat rheumatism.

In the Amazon, the leaves are used to treat malaria and watery jaundice from hepatitis to clean ulcers. It is antiabortive and astringent and is used to prevent the exhaustion of milk flow and counteract the clinical picture of cancer. In addition, it has antioxidant activity. It is also used to treat dysentery, bruises, and cuts. The rhizome tea is used to treat fever and

dyspepsia. The seed juice is used to relieve earache (Konnerup & Brir, 2010; Romero Vásquez & Valdez Valles, 2016; Flórez Londoño & Valencia Correa, 2019).

21.3 Other uses of achira plants

The achira plant is also used in animal nutrition and as an ornamental plant.

Starch can be used in the food and pharmaceutical industries since it is a substitute for conventional starches (see Chapter 14 in the First Volume of this series). It also has important characteristics, such as high viscosity of the gel, relatively low gelation temperature, and large granule size (Caicedo et al., 2000, 2003; González, 2012).

Recent studies have confirmed that achira fibers, like bamboo fibers, offer the possibility as natural fibers of using the material as a recycled raw material for the market (Gómez-Rosales et al., 2023).

21.3.1 Achira leaves and rhizomes as potential fodder for livestock in rural households

When achira is cultivated on large areas, for example, in Colombia, Peru, Ecuador, and Bolivia, this crop is part of familiar cropping systems. Smallholders practice intercropping and polyculture, and achira leaves can be a good alternative for fodder in rural households. Vaughan and Fernández (2013) determined above-ground canopy yield in three cultivars at 543–837 g fresh wt. · plant⁻¹, which equals 6.7–10.3 t · ha⁻¹ fresh wt. Nutrient content was also determined and ranged from 11.3% to 14.9% dry matter, 9.25%–11.0% protein on a dry weight basis. They also found that increasing leaf yield can reduce starch yield. However, it can be an additional

and practical source of fodder for small rural households in Guayata, Colombia.

Caguana (2017) tested four treatments (t1: 0%, t2: 8%, t3: 16%, and t4: 24% of flour) of feeding achira (*Canna edulis*) flour to 240 male guinea pigs (*Cavia porcellus*) at 8 weeks of age in Ambato, Ecuador. The most consumed diet was the second treatment with 70.29 g · day⁻¹, demonstrating that feeding this amount of achira flour can provide good productivity with better feed conversion.

Achira is a potential crop in the industry because it produces up to 10 t · ha⁻¹ of starch (see Chapter 14 in the Third Volume of this series). Achira starch has high viscosity compared to maize, potato, and cassava starch, which indicates that it has a high capacity to form and maintain stable gels (Piyachomkwan et al., 2002; Ribera Pilatasig, 2006; Srikaeo et al., 2011).

In the pharmaceutical industry achira starch is used as a coating for capsules and in the textile industry for finishing fabrics. It is also used in paper and packaging, in cosmetics, make-up, face creams, and in the adhesives industry (Bertolini, 2009; González, 2012).

21.4 Concluding remarks

The traditional uses, processes, and markets of achira (*C. indica*) present a rich tapestry of cultural heritage, ecological significance, and economic potential. Originating from South America, achira has traversed continents, finding its place in various cuisines, medicinal practices, and industrial applications worldwide.

Throughout history, achira has been revered for its multifaceted utility. From its roots in Incan gastronomy to its potential medicinal properties utilized across different cultures, achira embodies a plant deeply intertwined with human life. Its starch extraction process, handed down through generations, reflects a sustainable approach to agriculture, devoid of agrochemicals and in harmony with nature.

The versatility of achira extends beyond traditional uses. Its starch, with interesting physicochemical properties, could be an important ingredient in various industries, from food to pharmaceuticals. With its high starch yield, achira emerges as a promising crop for agroindustry, offering an alternative to conventional starch sources like maize and cassava.

One of the countries that has developed and progressed with processed product using achira is Colombia (Rodríguez et al., 2003). Today, this country has developed a traditional food based on achira starch named “Biscocho de Achira del Huila” (in English “Biscuit of Achira from the Huila” a region of Colombia) (La Patria, 2018), that is exported and is recognized by the European Union. Almidon de achira or achira starch has the potential to improve rural households, as confirmed in a study for a community in the canton of Loja, Ecuador—where in an economic feasibility study for 47,704 rural households, 70.7% responded that they would be willing to use it (Idrovo, 2010). Achira starch has potential as a commercial product, as demonstrated in the study by Linares-Falla and Rey-Novoa (2018), and could improve the incomes of rural households in Colombian achira producers. It is also recommended the inclusion of achira starch in popular baked and gastronomic products in Ecuador due to its easily digestibility and acceptance by consumers (Morejon, 2012).

Furthermore, achira’s phytochemical composition underscores its importance beyond mere sustenance. Rich in nutrients and bioactive compounds, achira exhibits pharmacological properties, contributing to its role in traditional medicine and wellness practices worldwide.

As we look to the future, achira symbolizes resilience and adaptability, bridging cultural traditions with modern innovations. Today it is an interesting and still neglected crop in Bolivia, but innovative restaurants such as Gustu are incorporating native potatoes,

achira, and other native products in fashion meals in a way to revalorize them (Zerda, de la, 2018; Luziatelli et al., 2023). Its continued cultivation and exploration hold promise for sustainable agriculture, economic development, and the preservation of cultural heritage. Embracing achira’s legacy offers a glimpse into the past and paves the way for a more sustainable and interconnected future.

References

- Alarcón, M., & Dufour, D. (1998). *Almidón agrio de yuca en Colombia: Tomo1: Producción y recomendaciones* (1st ed, p. 35) Cali: Centro Internacional de Agricultura Tropical (CIAT). http://ciat-library.ciat.cgiar.org/articulos_ciat/almidon_agrio_2.pdf.
- Andrade-Mahecha, M. M., Tapia-Blácido, D. R., & Menegalli, F. C. (2012). Physical–chemical, thermal, and functional properties of achira (*Canna indica* L.) flour and starch from different geographical origin. *Starch/Stärke*, 64, 348–358. Available from <https://doi.org/10.1002/star.201100149>.
- Ariza-León, E., Castro-Cely, A. L., & Gómez-Cañón, J. D. (2013). Factibilidad del uso del almidón de achira como agente controlador de filtrado en lodos de perforación base agua. *Revista ION*, 26(1), 63–71. Available from <http://www.scielo.org.co/scielo.php?script=sciarttext&pid=S0120-100X2013000100007>.
- Bertolini, A. (Ed.), (2009). *Starches: Characterization, properties, and applications* (1st ed.). CRC Press. Available from <https://doi.org/10.1201/9781420080247>.
- Burhannudin, N., Mahmudah, S., Widyarini, & Purnomosari, D. (2018). Chemopreventive effects of edible *Canna* (*Canna edulis* Kerr.) against colorectal carcinogenesis: Effects on expression of adenomatous *Polypsis coli* and inducible nitric oxide synthase in rat inflammatory model. *Asian Pacific Journal of Cancer Prevention*, 19 (3), 839–844. Available from <https://doi.org/10.22034/APJCP.2018.19.3.839>.
- Caguana, M. H. (2017). *Efecto de la achira (Canna edulis) sobre el consumo voluntario y la digestibilidad aparente de nutrientes en cuyes (Cavia porcellus) en la etapa de engorde* [MSc Thesis Med. Vet., Facultad de Ciencias Agropecuarias, Universidad Técnica de Ambato, Ambato-Tungurahua, Ecuador] (67 p.). <https://repositorio.uta.edu.ec/bitstream/123456789/25107/1/Tesis%2082%20Medicina%20Veterinaria%20y%20Zootecnia%20CD%20474.pdf>.
- Caguana Telenchana, V. E. (2018). *Evaluación de la Achira (Canna indica) para elaboración de Vajilla desechable*

- biodegradable [BSc Thesis, Universidad Técnica de Cotopaxi (UTC), Latacunga, Ecuador] (88 p.). <http://repositorio.utc.edu.ec/handle/27000/5238>.
- Caicedo, G. (2004). El cultivo de achira: Alternativa de producción para el pequeño productor. *Raíces Andinas: Contribuciones al conocimiento y a la capacitación* (pp. 149–156). Lima: CIP. Available from https://cipotato.org/wp-content/uploads/2014/06/11_El_cultivo_achira_alternat_produc.pdf.
- Caicedo, G., Rozo, L., & Bonilla, U. (2000). *La achira su producción y beneficio* (p. 51) Bogotá: CORPOICA, 8–10.
- Caicedo, G., Segundo, L., & Wilches, R. (2003). *Alternativa agroindustrial para áreas de economía campesina: Técnicas de cultivo, beneficio o proceso poscosecha y uso agroindustrial* (p. 86) Bogotá: CORPOICA. https://repository.agrosavia.co/bitstream/handle/20.500.12324/13024/Ver_Documento_13024.pdf?sequence=1&isAllowed=y. ISBN: 958-8210-4-5.
- Caicedo, G. E., Bravo, M., & Herrera, J. P. (2020). Evaluación de tres cultivares de achira (*Canna edulis* Ker) con periodo vegetativo corto y alta producción de almidón, para las zonas marginales de café en el Departamento del Huila. *Ingeniería y Región*, 24, 99–104. Available from <https://diainet.unirioja.es/descarga/articulo/8960918.pdf>.
- Cardenas, M. (1950). *Plantas alimenticias nativas de los Andes de Bolivia* (p. 25) Cochabamba: Universidad de Cochabamba.
- Cisneros, F. H., Zevillanos, R., & Cisneros-Zevallos, L. (2009). Characterization of starch from two ecotypes of Andean achira roots (*Canna edulis*). *Journal of Agricultural and Food Chemistry*, 57(16), 7363–7368. Available from <https://doi.org/10.1021/jf9004687>.
- Flórez Londoño, T., & Valencia Correa, J. (2019). *Análisis de crecimiento de Canna indica a partir de semillas sexuales sembradas en condiciones de invernadero de la Universidad del Quindío, Armenia Quindío* [Lic. Biol. & Educación Ambiental Thesis, Universidad del Quindío (UQ), Armenia, Quindío, Ecuador] (65 p.). <https://bdigital.uniquindio.edu.co/bitstream/handle/001/6052/AN%C3%81LISIS%20DE%20CRECIMIENTO%20DE%20Canna%20indica%20A%20PARTIR%20DE%20SEMILLAS%20SEXUALES%20SEMBRADAS%20EN%20CONDICIONES%20DE.pd.f?sequence=1&isAllowed=y>.
- Gade, D. W. (1966). Achira, the edible *Canna*, its cultivation and use in the Peruvian Andes. *Economic Botany*, 20, 407–415. Available from <https://doi.org/10.1007/BF02904063>.
- Gawl, K., Lobo-Arias, M., Medina-Cano, C. I., Grisales-Arias, J. D., Yepes-Agudelo, A. F., & Álvarez-Guzmán, J. A. (2017). Caracterización y evaluación morfológicas de la colección colombiana de achira, *Canna edulis* Ker Gawl. (Cannaceae) [Evaluation and morphological characterization of the Colombian collection of achira]. *Corpoica Ciencia y Tecnología Agropecuaria*, 18(1), 47–73. Available from https://doi.org/10.21930/rcta.vol18_num1_art:558.
- Gómez-Rosales, Z.-E., Rodrigo-Illarri, J., Castiblanco-Moncada, L.-J., Rodrigo-Clavero, M.-E., Solano-Meza, J.-K., & Orjuela-Yepes, D. (2023). Physical and thermal characterization of achira (*Canna edulis* Ker) fiber obtained from food industry waste in the Department of Cundinamarca, Colombia. *Applied Sciences*, 13(17), 9522. Available from <https://doi.org/10.3390/app13179522>.
- González, G. H. (2012). *Desarrollo de productos con alto contenido de almidón para la industria de alimentos* (pp. 1–51). Bogotá: Fundación Universitaria Agraria de Colombia-UNIAGRARIA.
- Hermann, H. (1994). *La achira y la arracacha: Procesamiento y desarrollo de nuevos productos* (pp. 10–12). Lima: CIP, Circular 9.
- Hermann, M., Quynh, N. K., & Peters, D. (1999). *Reappraisal of edible canna as a high-value starch crop in Vietnam*. CIP Program Report 1997–98: Impact on a changing world. International Potato Center Program Report 1997–1998 (pp. 415–424). Lima: CIP.
- Hernández Sierra, K. A. (2017). *Evaluación de propiedades de almidones de diferentes fuentes botánicas (achira, papa, yuca, maíz) y sus películas* [MSc thesis, Facultad de Ingeniería, Universidad de los Andes, Bogota, Colombia] (42 p.). <http://hdl.handle.net/1992/39608>.
- Idrovo, W. A. (2010). *Estudio de factibilidad para la producción y comercialización de harina de achira (Canna edulis) en el Canton Loja* [Tesis de Grado, Escuela Politécnica Nacional, Facultad de Ciencias Administrativas] (179 p.). <https://bibdigital.epn.edu.co/bitstream/15000/1292/1/CD-2663.pdf>.
- Imai, K. (2008). Edible canna: A prospective plant resource from South America. *Japanese Journal of Plant Science*, 2 (2), 46–53. Available from https://lavierebelle.org/IMG/pdf/canna_edible_potential.pdf.
- Kasemwong, K., Uttapap, D., Piyachomkwan, K., & Takeda, Y. (2003). A comparative study of edible canna (*Canna edulis*) starch from different cultivars. Part II. Molecular structure of amylose and amylopectin. *Carbohydrate Polymers*, 54(4), 489–498. Available from <https://doi.org/10.1016/j.carbpol.2003.08.003>.
- Konnerup, D., & Brir, H. (2010). Nitrogen nutrition of *Canna indica*: Effects of ammonium versus nitrate on growth, biomass allocation, photosynthesis, nitrate reductase activity and N uptake rates. *Aquatic Botany*, 92(2), 142–148. Available from <https://doi.org/10.1016/j.aquabot.2009.11.004>, Accessed 15.12.24.
- Lares, M., & Pérez, E. (2006). Determination of the mineral fraction and rheological properties of microwave modified starch from *Canna edulis*. *Plant Foods for Human Nutrition*, 61, 109–113. Available from <https://doi.org/10.1007/s11130-006-0007-7>.
- Linares-Falla, J., & Rey-Novoa, H. (2018). *Almidón de sagú (Canna edulis Ker), producto colombiano con potencial de*

- comercialización internacional. Available from: <https://hdl.handle.net/10983/22363>.
- Lobo-Arias, M., Medina-Cano, C. I., Grisales-Arias, J. D., Yepes-Agudelo, A. F., & Álvarez-Guzmán, J. A. (2016). Caracterización y evaluación morfológicas de la colección colombiana de achira, *Canna edulis* Ker Gawl. (Cannaceae). *Ciencia y Tecnología Agropecuaria*, 18(1), 47–73. Available from https://doi.org/10.21930/rcta.vol18_num1_art:558.
- Luziatelli, G., Alandia, G., Rodríguez, J. P., Manrique, I., Jacobsen, S. E., & Sørensen, M. (2023). Ethnobotany of Andean root crops: Tradition and innovation—Arracacha (*Arracacia xanthorrhiza* Bancr.), Yacón (*Smallanthus sonchifolius* (Poepp.) H. Rob.), Mauka (*Mirabilis expansa* (Ruíz & Pav.) Standl.), Ahipa (*Pachyrhizus ahipa* Parodi), Maca (*Lepidium meyenii* Walp.), Achira (*Canna indica* L.). *Varieties and landraces* (pp. 101–134). Academic Press.
- Montaldo, A. (1972). *Canna edulis*. *Cultivo de raíces y tubérculos tropicales* (pp. 201–203). Lima: Editorial IICA.
- Morales Ordoñez, K. E. (2016). *Propuesta de desarrollo de un recetario de postres a base de achira y rosas comestibles* [Pregrade Thesis, Facultad de Ciencias de la Hospitalidad, Universidad de Cuenca, Cuenca, Ecuador] (107 p.). <http://dspace.ucuenca.edu.ec/bitstream/123456789/24568/1/Trabajo%20de%20titulaci%3b%3n.pdf>.
- Morejon, K. E. (2012). *Utilización del almidón de achira en la elaboración de repostería ecuatoriana 2011* [MSc thesis, Facultad de Salud Publica, Escuela Superior Politécnica de Chimborazo, Riobamba, Ecuador] (82 p.). <http://dspace.esPOCH.edu.ec/bitstream/123456789/9470/1/84T00117.pdf>.
- Otari, S. V., Pawar, S. H., Patel, S. K. S., Singh, R. K., Kim, S.-Y., Lee, J. H., Zhang, L., & Lee, J.-K. (2017). *Canna edulis* leaf extract-mediated preparation of stabilized silver nanoparticles: Characterization, antimicrobial activity, and toxicity studies. *Journal of Microbiology and Biotechnology*, 27(4), 731–738. Available from <https://doi.org/10.4014/jmb.1610.10019>.
- La Patria. (2018, December 17). Unión Europea reconoció a nueve productos Colombianos con denominación. *Publicado en La Patria*. <<https://archivo.lapatria.com/nacional/union-europea-reconocio-nueve-productos-colombianos-con-denominacion-428692>>.
- Pazto Pasto, M. M., & Zaruma Zaruma, F. D. T. (2013). *Elaboración de galletas de sémola de trigo (Triticum aestivum vulgaris) con chispas de chocolate utilizando tres niveles de almidón de achira (Canna edulis) y tres porcentajes de manteca, en la planta de frutas y hortalizas de la universidad de Bolívar* [Ing. Agroindustrial Thesis, Escuela de Ingeniería Agroindustrial, Facultad de Ciencias Agropecuarias, Universidad Estatal de Bolívar (UEB), Guaranda, Ecuador] (108 p.). <https://dspace.ueb.edu.ec/bitstream/123456789/939/1/072.pdf>.
- Piyachomkwan, K., Chotineeranat, S., Kijkhunasatian, C., Tonwitawat, R., Prammanee, S., Oates, C. G., & Sriroth, K. (2002). Edible canna (*Canna edulis*) as a complementary starch source to cassava for the starch industry. *Industrial Crops and Products*, 16, 11–21. Available from <http://www.repository.rmutt.ac.th/dspace/bitstream/123456789/378/1/Editable%20canna%20%28Canna%20edulis%29%20as%20a%20complementary%20starch%20source%20to%20cassava%20for%20the%20st.pdf>.
- PoWo (Plants of the World Online). (2024). Board of Trustees of the Royal Botanic Gardens, Kew, Richmond, Surrey, UK. <https://powo.science.kew.org/>.
- Puncha-arnon, S., Puttanlek, C., Rungsardthong, V., Pathipanawat, W., & Uttapap, D. (2007). Changes in physicochemical properties and morphology of canna starches during rhizomal development. *Carbohydrate Polymers*, 70(2), 206–217. Available from <https://doi.org/10.1016/j.carbpol.2007.03.020>.
- Reichel, H. (2007). Partículas flexuosas de aspecto viral aisladas de achira (*Canna edulis* Ker.) afectada por clorosis en Colombia. *Agronomía Colombiana*, 25(2), 330–332. Available from <https://revistas.unal.edu.co/index.php/agrocol/article/view/14137/14892>.
- Ribera Pilatasig, J. H. (2006). *Extracción de almidón de achira (Canna edulis) y modificación por acetilación y doble reprivatización* [Ing. Alimentos Thesis, Facultad de Ciencia e Ingeniería en Alimentos, Universidad Técnica de Ambato, Ambato, Ecuador] (43 p.). <https://repositorio.uta.edu.ec/bitstream/123456789/3350/1/P86%20ref.2975.pdf>.
- Roca, W., & Manrique, I. (2005). Valorización de los recursos genéticos de raíces y tubérculos andinos para la nutrición y la salud. *Agrociencia*, 9(1–2), 195–201. Available from <https://beta.acuedi.org/book/5248/pdf>.
- Rodríguez, G. A., García, H. R., Camacho, J. H., & Arjas, F. L. (2003). *El almidón de achira o sagu (Canna edulis Ker.): Manual técnico para su elaboración* (p. 36) Tibaitata, Colombia: Corporación Colombiana de Investigación Agropecuaria (CORPOICA). https://repository.agrosavia.co/bitstream/handle/20.500.12324/13501/Ver_Documento_13501.pdf?sequence=1&isAllowed=y.
- Romero Vásquez, S. M., & Valdez Valles, J. A. (2016). *Capacidad antioxidante in vitro de los flavonoides totales del extracto fluido de las hojas de Canna indica L. (achira) frente al radical libre 2,2-difenil-1-picrilhidrazilo* [BSc Pharm. & Biochem. thesis, Universidad de Trujillo (UNT), Trujillo, Peru]. <https://hdl.handle.net/20.500.14414/4185>.
- Ruilova, H. (2013). *Influencia de la sustitución parcial de harina de trigo (Triticum aestivum L.) por almidón de achira (Canna edulis) en las características caloríficas, físico-químicas y organolépticas del cupcake* [BSc thesis, Ingeniería Agroindustrial, Universidad Estatal Amazónica (UEA), Puyo, Pastaza, Ecuador] (133 p.). <https://repositorio.uea>.

- edu.ec/xmlui/bitstream/handle/123456789/644/T.AGROIN.B.UEA.0022?sequence=3&isAllowed=y.
- Sarje, S. K., Ingole, K., Angad, S., Priya, B., & Ghiware, N. B. (2019). A pharmacognostic and pharmacological review on *Canna indica* Linn. *International Journal of Research in Pharmacy and Chemistry*, 9(3), 61–77. Available from <https://doi.org/10.33289/IJRPC.9.3.2019.929>.
- Srikaeo, K., Mingyai, S., & Sopade, P. A. (2011). Physicochemical properties, resistant starch content and enzymatic digestibility of unripe banana, edible canna, taro flours and their rice noodle products. *International Journal of Food Science & Technology*, 46(10), 2111–2117. Available from <https://doi.org/10.1111/j.1365-2621.2011.02724.x>.
- Tanaka, M., Koida, A., Miyazaki, A., Tabata, K., Takei, Y., Tanimoto, Y., Kawamura, M., Tsuzuki, M., Takahashi, H., Yano, T., & Watanabe, H. (2023). Canna starch improves immune functions and the intestinal environment in mice. *Bioscience of Microbiota, Food and Health*, 42(2), 131–137. Available from <https://doi.org/10.12938/bmfh.2022-068>.
- Tanaka, N. (2004). The utilization of edible Canna plants in southeastern Asia and southern China. *Economic Botany*, 58(1), 112–114. Available from [https://doi.org/10.1663/0013-0001\(2004\)058\[0112:NOEP\]2.0.CO;2](https://doi.org/10.1663/0013-0001(2004)058[0112:NOEP]2.0.CO;2).
- Tapia, M. E., & Fries, A. M. (2007). *Guía de campo de los cultivos Andinos* (p. 222) Lima: FAO & ANPE. <https://www.fao.org/3/ai185s/ai185s.pdf>. ISBN: 978-92-5-305682-8.
- Torres, L. J. (2004). *Tecnología para el cultivo del sagú o achira (Canna edulis Ker)* (p. 39) Bogota: Corpoica, Secretaría de Agricultura y Desarrollo Rural. https://repository.agrosavia.co/bitstream/handle/20.500.12324/2117/Ver_Documento_117.pdf?sequence=1&isAllowed=y. ISBN: 958-8210-56-9.
- Tropicos. (2016). *Missouri: Missouri Botanical Garden*. <http://www.tropicos.org> Accessed 15.01.24.
- Vaughan, G., & C. Fernández. (2013). El potencial forrajero de la achira o sagú (*Canna indica* L.) en Guayatá, Colombia. *Livestock Research for Rural Development* 25(10): Article #178. Accessed February 25 2024 at: <https://www.lrrd.cipav.org.co/lrrd25/10/vaug25178.html>
- WFO (World Flora Online). (2024). *The World Flora Online project*. <https://about.worldfloraonline.org/>.
- Yaruro Cáceres, N. C. (2018). *Evaluación de las propiedades fisicoquímicas, térmicas y microestructurales del almidón de achira (Canna edulis)* [MSc thesis, Facultad de Ciencias Agrarias, Universidad Nacional de Colombia (UNC), Bogota, Colombia] (133 p.). <https://repositorio.unal.edu.co/bitstream/handle/unal/69533/1143232250.2019.pdf?sequence=1&isAllowed=y>.
- de la Zerda, P. N. (2018, September 5). Los tesoros gastro-nómicos de Apolo. *La Razón*. La Paz, Bolivia. <https://www.la-razon.com/escape/2018/09/05/los-tesoros-gastronomicos-de-apolo/>.
- Zhang, J., Chen, F., Liu, F., & Wang, Z. W. (2010). Study on structural changes of microwave heat-moisture treated resistant *Canna edulis* Ker starch during digestion in vitro. *Food Hydrocolloids*, 24(1), 27–34. Available from <https://doi.org/10.1016/j.foodhyd.2009.07.005>.
- Zhang, J., Wang, Z. W., Yu, W. J., & Wu, J. H. (2011). Pectins from *Canna edulis* Ker residue and their physicochemical characterization. *Carbohydrate Polymers*, 83(1), 210–216. Available from <https://doi.org/10.1016/j.carbpol.2010.07.043>.